

Grade 8
Unit 3
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Complete the table and then answer question 1.

Number	Standard Scientific Form
4,500,000,000	4.5×10^9
45×10^6	4.5×10^7
45×10^7	4.5×10^8
45×10^5	4.5×10^6

1. Which numbers in the table are represented by the same standard scientific form?

None of them

2. Find the sum by filling in the blanks.

$$(5.07 \times 10^6) + (4.5 \times 10^5)$$

$$(\underline{50.7} \times 10^5) + (4.5 \times 10^5)$$

$$= (55.2 \times 10^5)$$

$$= (\underline{5.52} \times 10^6) \quad 5.52 \times 10^6$$

Use the given information to answer questions 3-5.

The population of Gainesville was 141,085 in 2020. The population of Jacksonville was 949,611 in 2020.

3. How many more people lived in Jacksonville compared to Gainesville in 2020? Write your answer in scientific notation.

$$808526 = 8.08526 \times 10^5$$

$$949611 - 141085 = 808526$$

4. Write the population numbers from the word problem in scientific notation.

141,085

$$1.41085 \times 10^5$$

949,611

$$9.49611 \times 10^5$$

5. What is the difference between the populations? Write your answer using scientific notation.

$$8.08526 \times 10^5$$

6. Two cells are viewed under a microscope. Cell A has a diameter of 4.02×10^{-6} and Cell B has a diameter of 6.8×10^{-5} . Fill in the blanks to find the difference in diameters of Cell A and Cell B.

$$(4.02 \times 10^{-6}) - (6.8 \times 10^{-5})$$

$$(.402 \times 10^{-5}) - (6.8 \times 10^{-5})$$

$$= (-6.398 \times 10^{-5})$$

$$= (-6.398 \times 10^{-5}) = -6.398 \times 10^{-5}$$

7. Find the difference. Write your answer using scientific notation.

$$\begin{aligned} &7.32 \times 10^5 - 9.68 \times 10^4 \\ &(7.32 \times 10^5) - (.968 \times 10^5) \\ &6.352 \times 10^5 \end{aligned}$$

8. Find the sum. Write your answer using scientific notation.

$$\begin{aligned} &4.2 \times 10^7 + 2.1 \times 10^5 \\ &4.2 \times 10^7 + .021 \times 10^7 = 4.221 \times 10^7 \end{aligned}$$

The table lists the average mass, in kilograms (kg) of different types of large sea creatures. Use the information to answer questions 9 and 10.

Creature	Mass of One Individual (kg)
Blue Whale	1.9×10^5
Orca	9.9×10^3
Grey Whale	2.3×10^3

9. What is the difference in the mass of a blue whale and a grey whale?

$$\begin{aligned} &1.9 \times 10^5 - 2.3 \times 10^3 \\ &190 \times 10^3 - 2.3 \times 10^3 \\ &= 187.7 \times 10^3 = 1.877 \times 10^5 \end{aligned}$$

10. What is the total mass of an orca and a blue whale?

$$\begin{aligned} &9.9 \times 10^3 + 1.9 \times 10^5 \\ &9.9 \times 10^3 + 190 \times 10^3 \\ &199.9 \times 10^3 = 1.999 \times 10^5 \end{aligned}$$